MIL-S-15191C
3 April 1986
SUPERSEDING
MIL-S-15191B
20 October 1967
(See 6.2)

MILITARY SPECIFICATION

SILICA, DIATOMACEOUS, UNCALCINED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of uncalcined diatomaceous silica pigment for use as a flatting extender pigment for coatings.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. The following specifications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-R-266 - Resin, Alkyd; Solutions.
TT-T-291 - Thinner, Paint, Mineral Spirits, Regular and Odorless.

MILITARY

DOD-P-15146 - Paint, Exterior, Dull Black (Formula No. 104). (Metric)
MIL-Z-15486 - Zinc Oxide, Technical (Acicular, Paint Use).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 8010
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2.1.2 Other Government document. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF LABOR
Code of Federal Regulations, Title 29
Part 1910, Section 1001 - Occupational Safety and Asbestos Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

(Copies of specifications and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 153 - Standard Test Methods for Specific Gravity of Pigments. (DoD adopted)
D 280 - Standard Test Methods for Hydroscopic Moisture (and Other Matter Volatile Under the Test Conditions) in Pigments. (DoD adopted)
D 281 - Standard Test Method for Oil Absorption of Pigments by Spatula Rub-Out.
D 476 - Standard Specification for Titanium Dioxide Pigments.
D 523 - Standard Test Method for Specular Gloss. (DoD adopted)
D 562 - Standard Test Method for Consistency of Paints Using the Stormer Viscometer. (DoD adopted)
D 717 - Standard Methods of Analysis of Magnesium Silicate Pigment. (DoD adopted)
D 1208 - Standard Test Methods for Common Properties of Certain Pigments. (DoD adopted)
D 1210 - Standard Test Method for Fineness of Dispersion of Pigment-Vehicle Systems. (DoD adopted)
D 1366 - Standard Practice for Reporting Particle Size Characteristics of Pigments.
D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)
E 11 - Standard Specification for Wire-Cloth Sieves for Testing Purposes. (DoD adopted)
E 97 - Standard Test Method for Directional Reflectance Factor, 45-deg, O-deg, of Opaque Specimens by Broad-Band Filter Reflectometry. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Diatomaceous silica shall be uncalcined silica having characteristic diatom structure. It shall be furnished as a dry pigment and shall be uniform and free from foreign grit or adulterants.

3.2 Particle shape and crystalinity. When examined under a petrographic microscope at a 300-power magnification, the diatomaceous silica shall show distinct diatom structures, including disk-like, spicular, cylindrical, and crescent-shaped forms or fragments thereof. Under polarized light, the pigment shall show substantially no birefringence, indicating the absence of crystalline matter.

3.3 Composition and properties. The diatomaceous silica and a test coating prepared as specified in 4.3.4 shall conform to the requirements shown in table I.

TABLE I. Requirements and test methods.

Component or property	Requirement	ASTM test method
Silica (SiO_2), minimum (percent mass)	85	D 717
Moisture and other volatile matter, maximum (percent mass)	6	D 280
Loss on ignition, maximum (percent mass)	6	D 1208
Specific gravity, maximum	2.1	D 153, method A
Dry brightness, minimum (percent reflectance)	60	E 97
Oil absorption, minimum/maximum (percent mass)	130 to 170	D 281
Asbestos	none present 1/	---
Coarse particles retained		
No. 50 sieve, maximum (percent mass)	none	E 11
No. 140 (106- m) sieve, maximum (percent mass)	0.5	E 11
No. 325 (45- m) sieve, maximum (percent mass)	3	E 11

See footnotes at end of table.

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TABLE I. Requirements and test methods.

Component or property	Requirement	ASTM test method
Fineness, minimum (Hegman units)	3	D 1210
Mean particle size, minimum/maximum (micrometers)	1.5 to 2.5	D 1366
Properties of test coating 2/		
Gloss, 60 degrees maximum	10	D 523
Consistency, Krebs-Stormer 200 r/min		
Grams	125 to 150	D 562
Krebs units (equivalent)	stated range 3/	D 562
Reflectance, maximum (percent)	stated value 3/	E 97
Color	none	none

1/ Test for asbestos shall be in accordance with the applicable OSHA test procedure.
2/ The test coating shall be prepared in accordance with 4.3.4.
3/ Requirements as stated in the coating specification or as stated in 3.3.1.

3.3.1 Where specific requirements are not listed in either table I or in the coating specification for which the pigment is to be used, the following values shall shall apply:

Property	Pigment requirements for coating specifications (see DOD-P-15146)
Properties of test coating	
Krebs units (equivalent)	67 to 72
Reflectance (percent)	20
Color	none

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Sampling. Two samples shall be taken at random from different packages from each lot, batch, day's pack, or other unit of production in a shipment. When no markings distinguishing between units of production appear, samples shall be taken from different packages in the ratio of two samples for each 10,000 pounds (4,540 kilograms (kg)), except that for shipments of less than 10,000 pounds two samples shall be taken. The samples may be tested separately or as a composite sample after blending in equal quantities the samples from the same production unit.

4.3 Tests. The tests to be used are listed in table I and in 3.2, 4.3.1, 4.3.2, 4.3.3, and 4.3.4.

4.3.1 Dry brightness. Compress a sample of the pigment in a dry powder press, and determine the reflectance of the smooth surface of the resultant disc in accordance with ASTM E 97.

4.3.2 Fineness. Weigh 78 grams of petroleum spirits in accordance with TT-T-291, type I and 80 grams of linseed oil (viscosity, 21 to 24 poise; acid number, 4 to 8) into a pint can and thoroughly mix with a spatula. Weigh in 160 grams of magnesium silicate and stir with a spatula to wet all lumps. Stir with a motor-driven propeller-type stirrer for 5 minutes at approximately 800 revolutions per minute (r/min). Weigh in an additional 180 grams of linseed oil and stir an additional 2 minutes with the electric stirrer. Determine the fineness in accordance with ASTM D 1210.

4.3.3 Mean particle size. Determine particle size distribution by a sedimentation method, record and plot the data according to ASTM D 1366. The diameter corresponding to the point on the curve that indicates that 50 percent of the mass of the pigment is of smaller size is the mean particle size.

4.3.4 Preparation of test coating. Prepare a mill base with the ingredients shown and reduce a weighed portion of the mill base with the additional ingredients shown. With the exception of the diatomaceous silica test pigment, the ingredients shall conform to the applicable specifications.

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Mill base

Ingredients	Parts by mass
Titanium dioxide (ASTM D 476, type II or III, containing a minimum of 88 percent TiO_2)	60
Zinc oxide (MIL-Z-15486)	210
Diatomaceous silica (test pigment)	230
Petroleum spirits (TT-T-291, type I)	50
Alkyd resin solution (TT-R-266, type I, class A)	465

Reduction

Additional ingredients	Parts by mass
Mill base paste	915
Petroleum spirits (TT-T-291, type I)	220
Drier 1/	---

1/ Type and amount of drier shall be the choice of the manufacturer. The drier shall contain no lead or other heavy metals and the paint shall dry within the specified time.

4.3.4.1 Thoroughly mix the mill base components, accurately weighed, in a change-can mixer for 0.5 hour. Pass the mixture once through a medium tightly set laboratory three-roller mill. In preparing the finished coating, reduce the weighed portion of the mill base with the remaining liquids and driers and mix thoroughly. One day after preparation of the coating, determine the properties listed in table I.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Packaging shall conform to ASTM D 3951.

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6. NOTES

6.1 Intended use. This specification is for a flatting extender pigment to be used in the manufacture of Government specification coatings and is not intended for direct acquisition by the Government.

6.2 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Air Force - 99
Navy - SH

Preparing activity:
Navy - SH
(Project 8010-1130)

Review activities:
Army - ME
Navy - AS
Air Force - 84

User activities:
Navy - YD, CG, OS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-15191C	SILICA, DIATOMACEOUS, UNCALCINED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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